

point

Honors Geometry • 1.1

line

Honors Geometry • 1.1

plane

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undefined terms

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collinear points

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coplanar points

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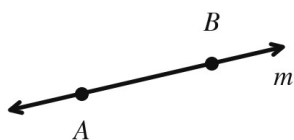
line segment

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ray

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A straight path that extends forever in both directions with no thickness. Named by two points on it or a lowercase letter.

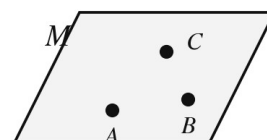


A location. It has no size — only position. Named with a capital letter and shown as a dot.

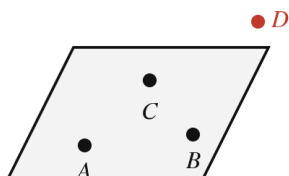


Point, line, and plane — the three words geometry starts with. They are described, not defined, and every other term is built from them.

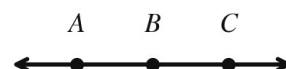
A flat surface that extends forever in every direction. Named by a capital letter or by three points in it that are not collinear.



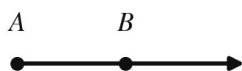
Points that lie in the same plane. (Any three points are coplanar; a fourth may not be.)



Points that lie on the same line.



Part of a line with ONE endpoint that extends forever in one direction. Named endpoint first: ray AB.



Part of a line made of two endpoints and every point between them. Written with a bar: $\overline{AB}$.



opposite rays

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intersection

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postulate (axiom)

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Ruler Postulate

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congruent segments

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between

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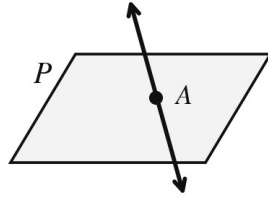
Segment Addition Postulate

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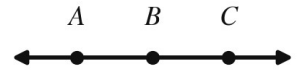
construction

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The set of points two figures have in common. Two lines meet in a point; two planes meet in a line.



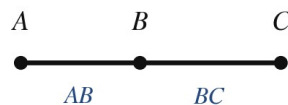
Two rays that share an endpoint and together form a line. Ray BA and ray BC are opposite rays if B is between A and C.



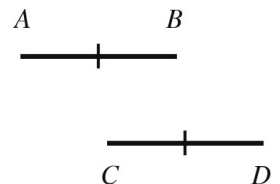
Every point on a line can be matched with a real number (its coordinate). The distance between two points is the absolute value of the difference of their coordinates.

A statement accepted as true without proof. The starting rules of geometry.

Point B is between A and C when all three are collinear and B lies on segment AC.

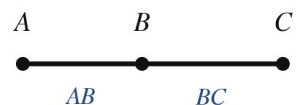


Segments that have the same length. Written $AB \cong CD$, and marked with matching tick marks.



A geometric drawing made using only a compass and a straightedge — no measuring.

If B is between A and C, then $AB + BC = AC$. (The parts add up to the whole.)



midpoint

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segment bisector

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bisect

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Midpoint Formula

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Distance Formula

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polygon

Honors Geometry • 1.4

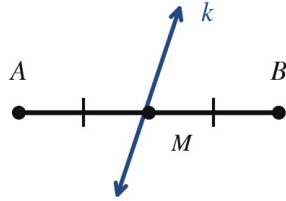
vertex (of a polygon)

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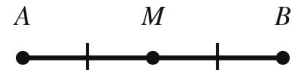
convex polygon

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A point, ray, line, line segment, or plane that intersects a segment at its midpoint.



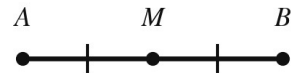
The point that divides a segment into two congruent segments.



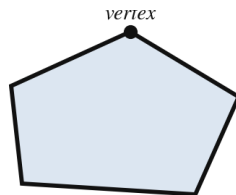
The midpoint of $A(x_1, y_1)$ and $B(x_2, y_2)$ is $M((x_1 + x_2)/2, (y_1 + y_2)/2)$.

Average the x's; average the y's.

To divide into two congruent parts.



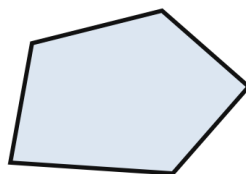
A closed plane figure made of three or more line segments that meet only at their endpoints. Each segment is a side.



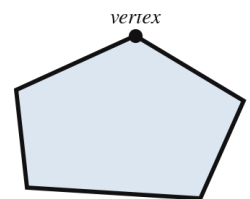
The distance between (x_1, y_1) and (x_2, y_2) is $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

It is the Pythagorean Theorem on a grid: $\sqrt{(\text{rise}^2 + \text{run}^2)}$.

A polygon in which NO line containing a side passes through the interior.



A point where two sides of a polygon meet. Plural: vertices.



concave polygon

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perimeter

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area

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angle

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sides of an angle

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acute angle

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right angle

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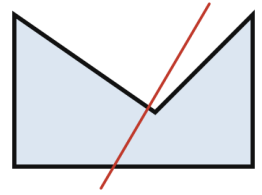
obtuse angle

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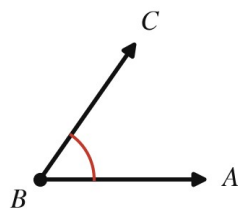
The distance around a figure — the sum of the side lengths. Measured in units (cm, ft, ...).



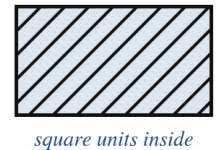
A polygon that is not convex — the line containing at least one side passes through the interior. It has a “dent.”



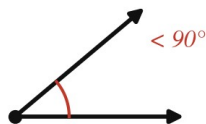
A figure formed by two rays with the same endpoint. The rays are the sides; the endpoint is the vertex. Named $\angle ABC$, $\angle CBA$, or $\angle B$.



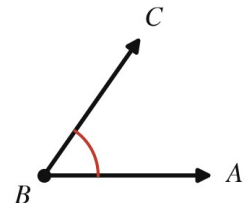
The amount of surface a figure covers — the number of square units inside it. Measured in square units (cm^2 , ft^2 , ...).



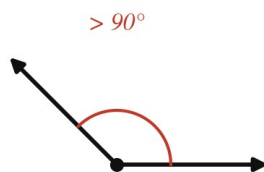
An angle whose measure is between 0° and 90° .



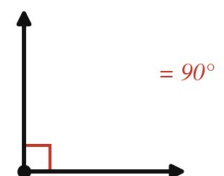
The two rays that form the angle. In $\angle ABC$, the sides are ray BA and ray BC.



An angle whose measure is between 90° and 180° .



An angle whose measure is exactly 90° . Marked with a small square at the vertex.



straight angle

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congruent angles

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angle bisector

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Angle Addition Postulate

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adjacent angles

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complementary angles

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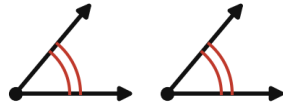
supplementary angles

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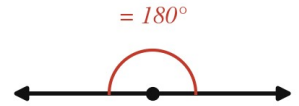
vertical angles

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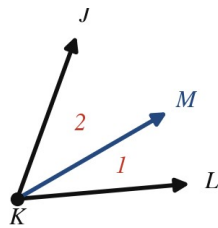
Angles that have the same measure. Written $\angle A \cong \angle B$, and marked with matching arcs.



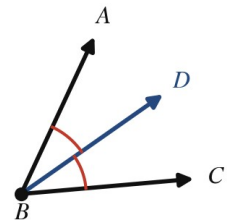
An angle whose measure is exactly 180° . Its sides are opposite rays, so it looks like a line.



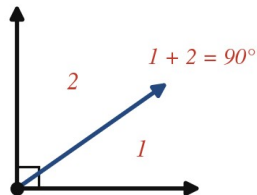
If M is in the interior of $\angle JKL$, then $m\angle JKM + m\angle MKL = m\angle JKL$. (The parts add up to the whole.)



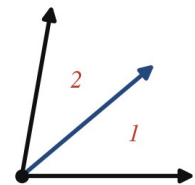
A ray that divides an angle into two congruent angles. If ray BD bisects $\angle ABC$, then $\angle ABD \cong \angle DBC$.



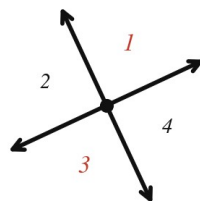
Two angles whose measures add to 90° . They may or may not be adjacent.



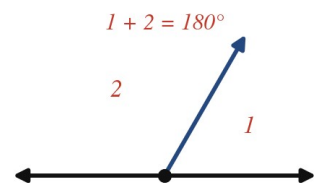
Two angles that share a common vertex and a common side but have no common interior points.



Two angles whose sides form two pairs of opposite rays — the “X” angles made by intersecting lines. Vertical angles are congruent.



Two angles whose measures add to 180° . They may or may not be adjacent.



linear pair

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Two adjacent angles whose noncommon sides are opposite rays. A linear pair is always supplementary.

